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SUMMARY

Computer-science student and applied AI engineer who builds multi-agent systems end-to-end — protocol design, security, and UI included. At **Tamarind Tech** I own and lead a client engagement building a **data-exposure monitoring tool** for a major bank in Myanmar. As personal projects, I designed and led the build of **Ammunity** (an AI-agent discovery and routing network with a from-scratch cryptographic agent-to-agent protocol) and built **Reckoner** (a comprehension gate for AI-assisted coding). Outside Tamarind Tech, I'm a core contributor to **vapviz**, an open-source AI-agent observability framework, and I build the data pipeline and six-model evaluation benchmark behind an academic facial-expression-recognition research paper. Comfortable across the stack: Python, TypeScript, applied cryptography and privacy engineering, computer vision, LLM tooling (CrewAI, Claude Code, Anthropic/OpenAI APIs), and cloud infrastructure (AWS, GCP, Docker).

EDUCATION

Siam University — Bangkok, Thailand

Bachelor of Information Technology | 2024 – Expected 2028

Relevant coursework: Programming, Data Structures, Computer Systems, Business Technology Solutions.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript / JavaScript, SQL
- **AI & LLM:** CrewAI, Anthropic Claude / Claude Code, OpenAI API, agent-to-agent protocol design, prompt engineering, LLM-as-judge evaluation, n8n
- **Computer Vision & ML:** OpenCV, ONNX (YuNet face detection, SFace face recognition), HSEmotion / EfficientNet emotion classification, face alignment (Umeyama transform, ArcFace template), image QC pipelines, PyTorch model benchmarking (Py-Feat, LibreFace, OpenFace 3.0, EmoNet, DDAMFN)
- **Evaluation & Statistics:** macro-F1, balanced accuracy, Cohen's κ , expected calibration error, bootstrap confidence intervals, McNemar tests (implemented in NumPy)
- **Security & Privacy:** Ed25519 signing, HMAC-SHA256 keyed hashing, k-anonymity lookups, challenge-response authentication, evidence integrity (atomic writes, hash checks), threat modelling
- **Backend:** FastAPI, Starlette, Flask / Jinja2 / htmx, Node.js, Ed25519 cryptographic signing (Python *cryptography*, Node *node:crypto*), Supabase (Postgres, Auth, RLS), SQLite (schema migrations), PostgreSQL, REST / WebSocket / Server-Sent Events
- **Frontend & Data Viz:** Next.js / React, ReactFlow, Zustand, Tailwind, canvas-based visualization
- **Infrastructure:** Docker, AWS (Lambda, Terraform, VPC networking, load balancers), GCP, Railway, Vercel, Linux, systemd, *uv* environment isolation
- **Data modelling:** knowledge graphs and ontology design, Neo4j graph data modelling (self-study)
- **Engineering practice:** test-driven development, mutation testing, directing multi-agent AI coding workflows with human approval gates, decision logs and threat models
- **Developer experience:** Git, GitHub, Bash, Claude Code, GitHub Copilot, ChatGPT

EXPERIENCE

AI Developer & Agent Architect — Tamarind Tech

Jan 2026 – Present | Focus: agent-to-agent communication and applied AI products

- Own and lead a client engagement for **a major bank in Myanmar**: took over an inherited data-exposure monitoring tool and am leading it through to client handover (see *Projects*).
- Within this role, conceived and designed **Ammunity** as a personal project — its cryptographic auth protocol, WebSocket delivery layer, and routing pipeline — and led its build, directing and reviewing AI coding agents (see *Projects*).
- Also independently built **Reckoner**, a comprehension gate for AI-assisted coding, as a personal project (see *Projects*).
- Led a security audit and hardening pass on Ammunity; maintains a public risk/flags ledger and closed ~45% of findings before controlled external testing.

Digital Marketing Associate — The Box Gallery, Bangkok

Aug 2024 – Apr 2025

- Executed paid-social campaigns and tracked performance in spreadsheet and SQL dashboards.
- Contributed to a ~50% increase in monthly store visits through audience-segmentation experiments.

PROJECTS

Data-Exposure Monitor — Customer-Data Protection for a Bank

Client engagement, Tamarind Tech — a major bank in Myanmar | Project Owner & Lead Engineer | Aug 2026 – Present

- Took ownership of an inherited defensive tool that works out which of a bank's customers have had their details exposed, so they can be warned, without sending customer data anywhere.
- Extended the layered detection engine (keyed-hash identity matching done entirely on-device, file-structure fingerprinting, bulk-PII heuristics) with two new layers for sensitive financial records, under a severity model that can only ever escalate.
- Rewrote and live-verified two third-party API integrations, making a privacy-preserving k-anonymity lookup (only 6 hash characters leave the machine) the default; added 3 new data sources.
- Fixed a core deduplication flaw that would have re-alerted on every hourly poll by separating “same subject” from “new information”; hardened evidence storage (atomic writes, fixed permissions, hash verification) and added run locking, alert retry and versioned SQLite schema migrations (2 → 15 tables).
- Built a local web review console that is structurally read-only against the engine's database, letting analysts confirm or dismiss each finding with a snapshot of exactly what was judged.
- Directed the build through a multi-agent AI coding workflow with human approval gates and independent testers: grew the engine suite from 44 to 1,600+ tests plus 650+ console tests, with mutation testing as the acceptance bar, and recorded 110+ scope and architecture decisions.

Python 3 · SQLite · Flask 3 / Jinja2 / htmx · HMAC-SHA256, SHA-256, SHA-1 k-anonymity · Claude Code multi-agent workflow

Ammunity — AI Agent Discovery & Routing Network

Personal project, built at Tamarind Tech | Architect & Lead Engineer | May 2026 – Present

- Designed a from-scratch agent-to-agent protocol (not Google's A2A) for mutually authenticated agent communication: a 5-layer permission gate followed by an Ed25519 challenge-response handshake (invite/accept/open/confirm) that binds every later message to a shared transcript hash for replay protection.
- Chose a “sign, don't encrypt” trust model so a human owner can audit their agent's conversations while the coordinator still cannot forge or silently alter a message — cross-language (Node signs, Python verifies) with pinned test vectors.

- Built the WebSocket delivery layer for agents with no public inbound endpoint; documented its single-instance scaling boundary and the Redis pub/sub fan-out required before horizontal scaling, rather than leaving it undiscovered.
- Designed a closed-by-default, community-scoped routing pipeline (3 sequential LLM calls — security check, discovery, hallucination-guarded capability match) at roughly \$0.0002/task, plus signed-reference file transfer over Supabase Storage with SHA-256 integrity checks.
- Backed by ~4,000 lines of Python tests and ~1,600 lines of Node tests; shipped through 8 documented protocol/schema version bumps (v0.3.0 → v0.7.1).

FastAPI / Starlette / Pydantic 2 / cryptography coordinator (Railway) · Node.js connector using built-in node:crypto Ed25519 (zero heavy dependencies) · Next.js 15 / React 19 / TypeScript / Tailwind web dashboard (Vercel) · Supabase Postgres with RLS on every table · built with AI coding agents under my direction and review · Live site: ammunity-web.vercel.app

vapviz — AI Agent Observability Framework

Open-source community project — originated by a collaborator, joined as core contributor Jun 2026, now co-maintainer | MIT license

- Joined an existing open-source AI-agent tracing framework after its initial release (built solo by a collaborator, Christopher Tun) and became primary owner of an entire new feature area: designed and shipped the project's 3-layer agent-control protocol (cooperative pause/resume/stop, message injection, and a full two-way ask/say conversation channel) across 4 published versions.
- Built the “Theater” and “Office Building” pixel-art monitoring views from scratch — a hand-authored sprite rasterizer, a room/floor animation engine, and tool-call-to-station routing — a novel way to watch many agents work across many apps on one screen.
- Fixed several correctness bugs inherited in the tracing core (a ContextVar state leak between test runs, unstable CrewAI parent-task nesting, incorrect LLM cost-table matching) and added the project's first real-LLM, non-mocked integration test suite.
- Owned the product/UX layer end-to-end: redesigned the raw technical graph into Simple / Technical / Story views, built the pixel design system, and shipped the public marketing site.

Python 3.11+ / FastAPI / SQLite (WAL) · React 18 / TypeScript / Vite / Zustand / ReactFlow + dagre · integrations for Anthropic, OpenAI, LangChain, CrewAI, Pydantic AI, LlamaIndex, AutoGen · GitHub: github.com/kochrisdev/vapviz · Live site: vapviz-website.vercel.app · current release v1.5.0

Reckoner — Comprehension Guardrails for AI-Assisted Coding

Personal project, built at Tamarind Tech | Jul 2026 | github.com/ARandomGuy9786/Reckoner

- Built a Claude Code PreToolUse hook that gates consequential actions (force-push, destructive migrations, prod deploys, secret edits) behind a “predict-then-reveal” comprehension check instead of a rubber-stamp approval.
- Designed a 3-tier resolver: free regex detectors, a library of pre-authored prediction “cards” validated by a Zod schema, and an on-the-fly LLM-generated card for novel actions under a persisted spawn budget.
- Solved the core constraint that a hook process has no terminal: invented a “deny-relay” protocol where the hook denies the call and instructs the agent to relay the question to the human, then re-verifies the answer against a fingerprinted, TTL-bound pending gate.
- Hardened the protocol against 3 real failure modes found in testing: an infinite re-ask loop when agents ran side commands mid-gate, a “spoiler” bug that leaked the answer before the prediction, and context loss when the relay stripped framing from the question.

TypeScript (strict, ESM, no build step via tsx) · @anthropic-ai/sdk · Zod-validated schemas shared between the CLI engine and the hook adapter

Myanmar Facial-Expression Dataset & Benchmark — Research Paper Contribution

Data & evaluation pipeline lead — academic research paper, outside research group | Jul 2026 – Present

- Own the data and evaluation pipeline behind a facial-expression-recognition (FER) research paper led by an outside academic research group, working with a supervising professor and teammates who carry out the human labelling. The result is a 378-face labelled dataset drawn from two Myanmar productions (a TV series and a feature film).
- **Stage 1 — candidate proposal:** samples a frame every 2 seconds (~188,000 frames from a 1h44m film in ~7 minutes), detects faces with YuNet and scores 6 Ekman emotions plus neutral with HSEmotion (EfficientNet), logging a full audit trail of every detection. Added a per-person cap (SFace face recognition, max 8 per identity per emotion), watermark-band exclusion, brightness and confidence floors set from measured distributions, and a clean-up pass for blur, poor exposure and near-duplicates.
- **Stage 2 — alignment:** deterministic (non-RANSAC) Umeyama alignment to an ArcFace template, producing lossless 224×224 colour and 48×48 grayscale images that are bit-identical and checksum-verified across runs. Deliberately excluded frontalization (it risks fabricating the expression being labelled), and removed CLAHE lighting correction after measuring that it washed out the film's dark cinematography.
- **Six-model evaluation kit:** benchmarks Py-Feat, LibreFace, OpenFace 3.0, EmoNet, DDAMFN and HSEmotion, each in an isolated *uv* environment to resolve conflicting PyTorch versions. Metrics are written from scratch in NumPy: macro-F1, balanced accuracy, Cohen's κ , calibration error, 2,000-sample bootstrap confidence intervals and McNemar significance tests. Verified that it reproduces the pipeline's own scores to within 5×10^{-4} .
- **Research integrity:** authored the data-collection and human-labelling protocols at the professor's request, including an appearance-based emotion rubric, a strict "no model ever assigns a label" rule, and an inter-annotator agreement and adjudication plan.

Python · OpenCV / NumPy / onnxruntime / ffmpeg · YuNet, SFace, HSEmotion · PyTorch (evaluation only) · uv · cross-platform (Windows / macOS / Linux) · every change reviewed via pull request

ADDITIONAL PROJECTS

- **AI-Powered Event-Based Travel Planner** — CrewAI multi-agent system that converts a budget and interest profile into a personalised itinerary; demoed live at a Microsoft-hosted showcase (Feb 2026).
- **OpenClaw ↔ Ammunity Integration Skill** — open-source Node.js skill letting a personal-assistant agent delegate to and receive from the Ammunity network, packaged as a systemd service.
- **n8n AI Assistant** — personal automation hub integrating the OpenAI API with n8n for reminders, daily summarisation, and inbox triage.

CERTIFICATIONS

AI / LLM — Anthropic

- Claude Code in Action — Anthropic (Mar 2026) [\[verify\]](#)
- Claude 101 — Anthropic (2026) [\[verify\]](#)
- Anthropic Claude certification — Apr 2026 [\[verify\]](#)
- Anthropic Claude certification — Apr 2026 [\[verify\]](#)
- Anthropic Claude certification — 2026 [\[verify\]](#)

Python — University of Michigan (Coursera)

- Python for Everybody Specialization (5 courses) — Jun 2025 [\[verify\]](#)

Data Analytics — Meta (Coursera)

- Introduction to Data Analytics — May 2025 [\[verify\]](#)
- Data Analysis with Spreadsheets and SQL — Jun 2025 [\[verify\]](#)

Networking

- Cisco Network Technician Career Badge

Digital Marketing

- Google Digital Marketing & E-commerce Professional Certificate — Oct 2024 [\[verify\]](#)
- Meta Social Media Marketing Professional Certificate — Mar 2025 [\[verify\]](#)
- Meta-Certified Digital Marketing Associate

LANGUAGES

English — Professional working proficiency · Burmese — Native